Phase 1 Report

Hypertension Risk Prediction Using Data Mining Techniques

1. Domain Context and Problem Understanding

1.1 Domain Overview

This project is situated in the **healthcare domain**, focusing on the prediction and analysis of hypertension risk. Hypertension, commonly known as high blood pressure, is a chronic medical condition that significantly increases the risk of cardiovascular diseases, stroke, kidney failure, and other serious health complications.

One of the major challenges associated with hypertension is that it often develops without noticeable symptoms, making early detection difficult. Therefore, identifying individuals at risk before the onset of severe complications is a critical task in preventive healthcare.

1.2 Problem Statement

The main objective of this project is to develop a **data mining-based system** capable of predicting whether a patient is at risk of hypertension using demographic, lifestyle, medical history, and clinical data.

1.3 Importance of the Problem

The significance of this problem lies in:

- enabling early detection of high-risk individuals
- supporting preventive healthcare strategies
- reducing long-term healthcare costs
- improving patient awareness and decision-making

1.4 Data Mining Questions

This project aims to answer the following questions:

- Can hypertension risk be accurately predicted using patient data?
- Which features contribute most to hypertension risk?
- What patterns exist between lifestyle, medical history, and hypertension?
- How can data-driven insights support healthcare decision-making?

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2. Dataset Description and Preview

2.1 Dataset Overview

The dataset used in this study contains:

- **10,000 patient records**
- **25 attributes (features)**

It includes a diverse set of variables related to patient demographics, lifestyle, medical history, and clinical measurements.

2.2 Feature Categories

Demographic Features

- age
- sex
- residence

Lifestyle Features

- smoking
- alcohol_heavy
- physically_active
- high_salt_diet

Medical History Features

- family_history_hypertension
- diabetes
- stroke_history
- myocardial_infarction
- heart_failure

Clinical Measurements

- bmi
- total_cholesterol_mg_dl
- ldl_mg_dl
- hdl_mg_dl
- creatinine_mg_dl

2.3 Target Variable

- `hypertension_status`

This variable represents whether a patient is:

- **Hypertensive (1)**
- **Normal (0)**

2.4 Data Types

The dataset consists of:

- Numerical variables (e.g., age, BMI, cholesterol levels)
- Categorical variables (e.g., sex, residence)
- Boolean variables (e.g., smoking, physically_active)

2.5 Data Quality Issues

Several data quality issues were identified during initial exploration:

1. Missing Values

- The column `medication_adherence` contains missing values

2. Irrelevant Features

- `patient_id` is an identifier and does not contribute to prediction

3. Data Leakage Risk

Some features directly reveal the target or strongly correlate with it:

- systolic_bp_mmhg
- diastolic_bp_mmhg
- diagnosed
- on_medication
- bp_controlled
- medication_adherence

These variables will be removed during preprocessing to ensure model validity.

2.6 Outlier Analysis

Outliers were detected using the **Interquartile Range (IQR) method**. Several features such as BMI, cholesterol levels, and creatinine showed a small number of extreme values.

However, given the healthcare nature of the dataset, these values are likely to represent real clinical conditions rather than errors. Therefore, instead of removing them, a **capping (Winsorization) approach** will be applied.

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3. CRISP-DM Plan

This project follows the **CRISP-DM (Cross-Industry Standard Process for Data Mining)** methodology.

3.1 Business Understanding

- Define the healthcare problem
- Understand the importance of hypertension prediction
- Set objectives and evaluation criteria

3.2 Data Understanding

- Load and inspect dataset
- Analyze structure and feature types
- Identify missing values and duplicates
- Perform initial visual exploration

3.3 Data Preparation

- Remove irrelevant and leakage features
- Clean and encode the target variable
- Encode categorical variables
- Handle missing values
- Detect and treat outliers
- Prepare final dataset for modeling

3.4 Modeling

- Apply classification algorithms:
  - Logistic Regression
  - Decision Tree
  - Random Forest
- Perform feature selection
- Optimize model performance

3.5 Evaluation

- Evaluate models using:
  - Accuracy
  - Precision
  - Recall
  - F1-score
  - Confusion matrix
- Analyze model reliability

3.6 Deployment

- Provide user-friendly interface
- Visualize predictions and insights

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4. Technique Selection Rationale

4.1 Classification

The primary task is binary classification since the goal is to predict whether a patient is hypertensive or not.

4.2 Exploratory Data Analysis (EDA)

EDA is used to:

- understand feature distributions
- detect outliers
- analyze relationships
- guide preprocessing decisions

4.3 Feature Selection

Feature selection helps:

- improve model performance
- reduce noise
- enhance interpretability

4.4 Optional Extension: Clustering

Clustering may be applied to identify patient groups with similar characteristics, adding a deeper data mining perspective.

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5. Environment and Repository Setup

A GitHub repository has been created with:

- structured folders (data, notebooks, src, reports, slides)
- README file describing the project
- .gitignore file for Python
- MIT License

The project environment is based on Python with libraries such as:

- pandas
- numpy
- matplotlib
- seaborn
- scikit-learn
